

Robin Walters

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Research on the role of symmetry in deep learning: how to build it in, when to relax it, and what it does to learning. Current threads: equivariant and relaxed-symmetry policies for robot manipulation and physical dynamics; parameter-space symmetry and its effect on optimization; symmetry and governing-equation discovery from data; and the theory of symmetry in functions and distributions.

APPOINTMENTS

2022–present	Assistant Professor , Khoury College of Computer Sciences, Northeastern University Boston, MA
2024–2025	Visiting Faculty Fellow , Robotics and AI Institute (formerly Boston Dynamics AI Institute) Cambridge, MA
2020–2022	Postdoctoral Research Fellow , Institute for Experiential AI and Roux Institute, Khoury College, Northeastern University Boston, MA
2015–2020	Zelevinsky Research Instructor and NSF Postdoctoral Teaching Associate, Department of Mathematics, Northeastern University Boston, MA
2008–2009	Technical Director (Resident) , Pixar Animation Studios Emeryville, CA · 3D rendering for <i>Toy Story</i> 1 and 2

EDUCATION

2009–2015	Ph.D. in Mathematics , University of Chicago Advisor: Victor Ginzburg
2004–2008	A.B. in Mathematics with High Honors , Harvard University

HONORS AND AWARDS

2026	Khoury College Junior Researcher Award Awarded to one pre-tenure faculty member per year
2026	Honorable Mention, IEEE Robotics and Automation Letters (RA-L) Five of 1,700 papers
2024	Outstanding Paper Finalist, Conference on Robot Learning (CoRL)
2024	Best Paper, 2nd Workshop on High-dimensional Learning Dynamics (HiLD), ICLR
2015–2019	NSF Mathematical Sciences Postdoctoral Research Fellowship Representation theory, with Valerio Toledano Laredo, Northeastern University
2010–2014	NSF Graduate Research Fellowship
2012	Wayne C. Booth Prize for Excellence in Teaching, University of Chicago Four graduate students university-wide each year

RESEARCH FUNDING

Period	Award	Role	Amount
2025–2030	NSF CAREER: Improving Data Efficiency in Deep Learning with Relaxed Symmetry Constraints	PI	\$600,000
2023–2027	NSF FRR: Symmetric Policy Learning for Robotic Manipulation (<i>PI: Robert Platt</i>)	Co-PI	\$866,735
2022–2025	NSF SCALE MoDL: Representation Theoretic Foundations of Deep Learning (<i>with J.-W. van de Meent; R. Yu, UCSD</i>)	Lead PI (NEU)	\$661,238 +\$300k UCSD
2022–2026	NSF RI: Medium: Learning MDP Abstractions for Autonomous Systems using Variational Methods and Symmetry Groups (<i>PI: Lawson Wong</i>)	Co-PI	\$1,200,000
2022–2026	Deep Equivariant Radar Models (DE-RM), ICE Line — U.S. Department of the Air Force through MIT Lincoln Laboratory	PI	\$340,000
2024–2025	Northeastern TIER 1 Interdisciplinary Seed Grant (<i>with Peter Schindler, MIE</i>)	Co-PI	\$50,000
2024–2025	Institute for Experiential AI Seed Funding (<i>with Peter Schindler, MIE</i>)	Co-PI	\$60,000
2026–2028	Renaissance Philanthropy : Symmetry-Aware Generative AI for Differential Equations and Mathematical Discovery (<i>with C. Garraffo, CfA; W. Freeman, MIT; K. He, MIT; M. Weber, Harvard</i>)	PI (NEU)	\$189,217 +\$799k others
2026–2029	NSF CAIG: HydrotherML: Machine Learning Accelerated Hydrothermal Modeling For Earth Science (<i>with N. Randolph-Flagg, UKY</i>)	PI (NEU)	\$350,000 +\$350k UKY
Total (Northeastern share, as PI or co-PI)			\$4,317,190

PUBLICATIONS

* equal contribution; † equal advising; mathematics publications list authors alphabetically.

Selected Publications

- [S1] Mingxi Jia, Haojie Huang, Zhewen Zhang, Chenghao Wang, Linfeng Zhao, Dian Wang, Jason Xinyu Liu, **Robin Walters**, Robert Platt, and Stefanie Tellex. Learning Efficient and Robust Language-conditioned Manipulation using Textual-Visual Relevancy and Equivariant Language Mapping. *IEEE Robotics and Automation Letters (RA-L)*, 2025. **Honorable Mention**.
RA-L Honorable Mention (5 of 1700 papers).
- [S2] Bo Zhao, **Robin Walters**, and Rose Yu. Symmetry in Neural Network Parameter Spaces. *Transactions on Machine Learning Research (TMLR)*, 2025.
Survey of symmetry in neural network parameter spaces and its consequences for training.
- [S3] Dian Wang, Stephen Hart, David Surovik, Tarik Kelestemur, Haojie Huang, Haibo Zhao, Mark Yeatman, Jiuguang Wang, **Robin Walters**, and Robert Platt. Equivariant Diffusion Policy. *Conference on Robot Learning (CoRL)*, 2024. **Outstanding Paper Finalist**.
Symmetry-constrained diffusion policies for robot manipulation; extended journal version in IJRR (2026). Outstanding Paper Finalist (6 out of 617 submissions).
- [S4] Bo Zhao, Robert M. Gower, **Robin Walters**, and Rose Yu. Improving Convergence and Generalization Using Parameter Symmetries. *International Conference on Learning Representations (ICLR)*, 2024. **Oral**.
Oral. (86 out of 7304 submissions).
- [S5] Derek Lim, Moe Putterman, **Robin Walters**, Haggai Maron, and Stefanie Jegelka. The Empirical Impact of Neural Parameter Symmetries, or Lack Thereof. *Neural Information Processing Systems (NeurIPS)*, 2024. **Best Paper, Workshop on High-dimensional Learning Dynamics (HiLD), ICLR 2024**.
Isolates the effect of neural parameter symmetries by constructing networks that lack them.

- [S6] David Klee, Ondrej Biza, Robert Platt, and **Robin Walters**. Image to Sphere: Learning Equivariant Features for Efficient Pose Prediction. *International Conference on Learning Representations (ICLR)*, 2023. **Oral, notable top 5%**.
SO(3)-equivariant object pose prediction from a single image by projecting features to the sphere. Oral (91 out of 4955 submissions.)
- [S7] Dian Wang, Jung Yeon Park, Neel Sortur, Lawson L.S. Wong, Robert Platt[†], and **Robin Walters**[†]. The Surprising Effectiveness of Equivariant Models in Domains with Latent Symmetry. *International Conference on Learning Representations (ICLR)*, 2023. **Spotlight, notable top 25%**.
Shows equivariant models remain effective when symmetry holds only in a latent space.
- [S8] Ayan Chatterjee, **Robin Walters**, Zohair Shafi, Omair Shafi Ahmed, Michael Sebek, Deisy Gysi, Rose Yu, Tina Eliassi-Rad, Albert-László Barabási, and Giulia Menichetti. Improving the generalizability of protein-ligand binding predictions with AI-Bind. *Nature Communications*, 14(1):1989, 2023.
Generalizable protein-ligand binding prediction that addresses shortcut learning in prior models.
- [S9] Dian Wang, Xupeng Zhu, Jung Yeon Park, Robert Platt, and **Robin Walters**. A General Theory of Correct, Incorrect, and Extrinsic Equivariance. *Neural Information Processing Systems (NeurIPS)*, 2023.
Error bounds for models whose assumed symmetry does not match the symmetry of the data.
- [S10] Rui Wang, **Robin Walters**, and Rose Yu. Approximately Equivariant Networks for Imperfectly Symmetric Dynamics. *International Conference on Machine Learning (ICML)*, 2022.
Relaxes exact equivariance for dynamical systems that are only approximately symmetric.
- [S11] Rui Wang*, **Robin Walters***, and Rose Yu. Incorporating Symmetry into Deep Dynamics Models for Improved Generalization. *International Conference on Learning Representations (ICLR)*, 2021.
Equivariant dynamics models for physical forecasting, including turbulent flow and ocean currents.
- [S12] Nima Dehmamy, **Robin Walters**, Liu Yanchen, Dashun Wang, and Rose Yu. Automatic Symmetry Discovery with Lie Algebra Convolutional Network. *Neural Information Processing Systems (NeurIPS)*, 2021.
Learns the Lie algebra of a dataset's continuous symmetry group directly from data.

Preprints and Manuscripts Under Review

- [P1] Claire Schlesinger, Circe Hsu, Peter Schindler, and **Robin Walters**. PRISMat: Policy-Driven, Permutation-Invariant Autoregressive Material Generation. arXiv:2605.16612, 2026.
- [P2] Omkar Patil, Ondrej Biza, Thomas Weng, Karl Schmeckpeper, Wil Thomason, Xiaohan Zhang, **Robin Walters**, Nakul Gopalan, Sebastian Castro, and Eric Rosen. You've Got a Golden Ticket: Improving Generative Robot Policies With A Single Noise Vector. arXiv:2603.15757, 2026.
- [P3] Owen Howell, Linfeng Zhao, Xupeng Zhu, Yaoyao Qian, Haojie Huang, Lingfeng Sun, Wil Thomason, Robert Platt, and **Robin Walters**. Clebsch-Gordan Transformer: Fast and Global Equivariant Attention. arXiv:2509.24093, 2025.
- [P4] Jung Yeon Park, Yuxuan Chen, Floor Eijkelboom, Jan-Willem van de Meent, Lawson L.S. Wong, and **Robin Walters**. Discovering Lie Groups with Flow Matching. arXiv:2512.20043, 2025.
- [P5] Marco Pacini, Mircea Petrache, Bruno Lepri, Shubhendu Trivedi, and **Robin Walters**. On Universality of Deep Equivariant Networks. arXiv:2510.15814, 2025.
- [P6] Neel Sortur, Justin Goodwin, Purvik Patel, Luis Enrique Martinez, Jr., Tzofi Klinghoffer, Rajmonda S. Caceres, and **Robin Walters**. Radar2Shape: 3D Shape Reconstruction from High-Frequency Radar using Multiresolution Signed Distance Functions. arXiv:2511.17484, 2025.
- [P7] Jordan Ganev and **Robin Walters**. Quiver neural networks. arXiv:2207.12773, 2022.
- [P8] Jordan Ganev, Twan van Laarhoven, and **Robin Walters**. Universal approximation and model compression for radial neural networks. arXiv:2107.02550, 2021.

Peer-Reviewed Conference Publications

- [C1] Saumya Gupta, Scott Biggs, Moritz Laber, Zohair Shafi, **Robin Walters**, and Ayan Paul. DeepWeightFlow: Re-Basined Flow Matching for Generating Neural Network Weights. *International Conference on Learning Representations (ICLR)*, 2026.
- [C2] Xupeng Zhu, Yu Qi, Yizhe Zhu, **Robin Walters**, and Robert Platt. EquAct: An SE(3)-Equivariant Multi-Task Transformer for Open-Loop Robotic Manipulation. *International Conference on Learning Representations (ICLR)*, 2026.
- [C3] David Klee, Boce Hu, Andrew Cole, Heng Tian, Dian Wang, Robert Platt, and **Robin Walters**. RAVEN: End-to-end Equivariant Robot Learning with RGB Cameras. *International Conference on Learning Representations (ICLR)*, 2026.
- [C4] Hannah Lawrence, Elyssa Hofgard, Vasco Portilheiro, Yuxuan Chen, Tess Smidt, and **Robin Walters**. To Augment or Not to Augment? Diagnosing Distributional Symmetry Breaking. *International Conference on Learning Representations (ICLR)*, 2026.
- [C5] Jung Yeon Park, Yuxuan Chen, Floor Eijkelboom, Jan-Willem van de Meent, Lawson L.S. Wong, and **Robin Walters**. Discovering Symmetry Groups with Flow Matching. *International Conference on Machine Learning (ICML)*, 2026.
- [C6] Edward Berman, Luisa Li, Jung Yeon Park, and **Robin Walters**. Smoothness Errors in Dynamics Models and How to Avoid Them. *International Conference on Machine Learning (ICML)*, 2026.
- [C7] Lakshita Dodeja, Ondrej Biza, Shivam Vats, Stephen Hart, Stefanie Tellex, **Robin Walters**, Karl Schmeckpeper, and Thomas Weng. When Life Gives You BC, Make Q-functions: Extracting Q-values from Behavior Cloning for On-Robot Reinforcement Learning. *Robotics: Science and Systems (RSS)*, 2026.
- [C8] Nathan Vaska, Justin Goodwin, **Robin Walters**, and Rajmonda Caceres. Reducing the Sensitivity of Neural Physics Simulators to Mesh Topology via Pretraining. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.
- [C9] Jung Yeon Park, Sujay Bhatt, Sihan Zeng, Lawson L.S. Wong, Alec Koppel, Sumitra Ganesh, and **Robin Walters**. Approximate Equivariance in Reinforcement Learning. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025.
- [C10] Manu Bhat, Jonghyun Park, Jianke Yang, Nima Dehmamy, **Robin Walters**, and Rose Yu. AtlasD: Automatic Local Symmetry Discovery. *International Conference on Machine Learning (ICML)*, 2025.
- [C11] Haibo Zhao, Dian Wang, Yizhe Zhu, Xupeng Zhu, Owen Howell, Linfeng Zhao, Yaoyao Qian, **Robin Walters**, and Robert Platt. Hierarchical equivariant policy via frame transfer. *International Conference on Machine Learning (ICML)*, 2025.
- [C12] Xupeng Zhu, Fan Wang, **Robin Walters**, and Jane Shi. SE(3)-Equivariant Diffusion Policy in Spherical Fourier Space. *International Conference on Machine Learning (ICML)*, 2025.
- [C13] Bo Zhao, Nima Dehmamy, **Robin Walters**, and Rose Yu. Understanding Mode Connectivity via Parameter Space Symmetry. *International Conference on Machine Learning (ICML)*, 2025.
- [C14] Boce Hu, Dian Wang, David Klee, Heng Tian, Xupeng Zhu, Haojie Huang, Robert Platt, and **Robin Walters**. 3D Equivariant Visuomotor Policy Learning via Spherical Projection. *Neural Information Processing Systems (NeurIPS)*, 2025.
- [C15] Dian Wang, Boce Hu, Shuran Song, **Robin Walters**[†], and Robert Platt[†]. A Practical Guide for Incorporating Symmetry in Diffusion Policy. *Neural Information Processing Systems (NeurIPS)*, 2025.
- [C16] Colin Kohler, Purvik Patel, Nathan Vaska, Justin Goodwin, Matthew C Jones, Robert Platt, Rajmonda S. Caceres, and **Robin Walters**. Bridging Equivariant GNNs and Spherical CNNs for Structured Physical Domains. *Neural Information Processing Systems (NeurIPS)*, 2025.

- [C17] Linfeng Zhao, Owen Howell, Xupeng Zhu, Jung Yeon Park, Zhewen Zhang, **Robin Walters**, and Lawson L.S. Wong. Equivariant Action Sampling for Reinforcement Learning and Planning. *Workshop on the Algorithmic Foundations of Robotics (WAFR)*, 2025.
- [C18] Dian Wang, Stephen Hart, David Surovik, Tarik Kelestemur, Haojie Huang, Haibo Zhao, Mark Yeatman, Jiuguang Wang, **Robin Walters**, and Robert Platt. Equivariant Diffusion Policy. *Conference on Robot Learning (CoRL)*, 2024. **Outstanding Paper Finalist**.
- [C19] Haojie Huang, Karl Schmeckpeper, Dian Wang, Ondrej Biza, Yaoyao Qian, Haotian Liu, Mingxi Jia, Robert Platt, and **Robin Walters**. Imagination Policy: Using Generative Point Cloud Models for Learning Manipulation Policies. *Conference on Robot Learning (CoRL)*, 2024.
- [C20] Boce Hu, Xupeng Zhu, Dian Wang, Zihao Dong, Haojie Huang, Chenghao Wang, **Robin Walters**, and Robert Platt. OrbitGrasp: SE(3)-Equivariant Grasp Learning. *Conference on Robot Learning (CoRL)*, 2024.
- [C21] Haojie Huang, Haotian Liu, Dian Wang, **Robin Walters**[†], and Robert Platt[†]. Match Policy: A Simple Pipeline from Point Cloud Registration to Manipulation Policies. *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.
- [C22] Haojie Huang, Owen Howell, Xupeng Zhu, Dian Wang, **Robin Walters**, and Robert Platt. Fourier Transporter: Bi-Equivariant Robotic Manipulation in 3D. *International Conference on Learning Representations (ICLR)*, 2024.
- [C23] Bo Zhao, Robert M. Gower, **Robin Walters**, and Rose Yu. Improving Convergence and Generalization Using Parameter Symmetries. *International Conference on Learning Representations (ICLR)*, 2024. **Oral**.
- [C24] Rui Wang, Elyssa Hofgard, Han Gao, **Robin Walters**, and Tess Smidt. Discovering Symmetry Breaking in Physical Systems with Relaxed Group Convolution. *International Conference on Machine Learning (ICML)*, 2024.
- [C25] Jianke Yang, Nima Dehmamy, **Robin Walters**, and Rose Yu. Latent Space Symmetry Discovery. *International Conference on Machine Learning (ICML)*, 2024.
- [C26] Lucas Laird, Circe Hsu, Asilata Bapat, and **Robin Walters**. MatrixNet: Learning over symmetry groups using learned group representations. *Neural Information Processing Systems (NeurIPS)*, 2024.
- [C27] Jianke Yang, Wang Rao, Nima Dehmamy, **Robin Walters**, and Rose Yu. Symmetry-Informed Governing Equation Discovery. *Neural Information Processing Systems (NeurIPS)*, 2024.
- [C28] Derek Lim, Moe Putterman, **Robin Walters**, Haggai Maron, and Stefanie Jegelka. The Empirical Impact of Neural Parameter Symmetries, or Lack Thereof. *Neural Information Processing Systems (NeurIPS)*, 2024. **Best Paper, Workshop on High-dimensional Learning Dynamics (HiLD), ICLR 2024**.
- [C29] Ondrej Biza, Skye Thompson, Kishore Reddy Pagidi, Abhinav Kumar, Elise van der Pol, **Robin Walters**, Thomas Kipf, Jan-Willem van de Meent, Lawson L.S. Wong, and Robert Platt. One-shot Imitation Learning via Interaction Warping. *Conference on Robot Learning (CoRL)*, 2023.
- [C30] Haojie Huang, Dian Wang, Xupeng Zhu, **Robin Walters**, and Robert Platt. Edge Grasp Network: A Graph-Based SE(3)-invariant Approach to Grasp Detection. *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.
- [C31] Mingxi Jia, Dian Wang, Guanang Su, David Klee, Xupeng Zhu, **Robin Walters**, and Robert Platt. SEIL: Simulation-augmented Equivariant Imitation Learning. *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.
- [C32] David Klee, Ondrej Biza, Robert Platt, and **Robin Walters**. Image to Sphere: Learning Equivariant Features for Efficient Pose Prediction. *International Conference on Learning Representations (ICLR)*, 2023. **Oral, notable top 5%**.
- [C33] Linfeng Zhao, Xupeng Zhu, Lingzhi Kong, **Robin Walters**, and Lawson L.S. Wong. Integrating

- Symmetry into Differentiable Planning with Steerable Convolutions. *International Conference on Learning Representations (ICLR)*, 2023.
- [C34] Bo Zhao, Iordan Ganev, **Robin Walters**, Rose Yu, and Nima Dehmamy. Symmetries, flat minima, and the conserved quantities of gradient flow. *International Conference on Learning Representations (ICLR)*, 2023.
- [C35] Dian Wang, Jung Yeon Park, Neel Sortur, Lawson L.S. Wong, Robert Platt[†], and **Robin Walters**[†]. The Surprising Effectiveness of Equivariant Models in Domains with Latent Symmetry. *International Conference on Learning Representations (ICLR)*, 2023. **Spotlight, notable top 25%**.
- [C36] Jianke Yang, **Robin Walters**, Nima Dehmamy, and Rose Yu. Generative Adversarial Symmetry Discovery. *International Conference on Machine Learning (ICML)*, 2023.
- [C37] Sophia Huiwen Sun, **Robin Walters**, Jinxi Li, and Rose Yu. Probabilistic Symmetry for Multi-Agent Dynamics. *Learning for Dynamics and Control (L4DC)*, 2023.
- [C38] Dian Wang, Xupeng Zhu, Jung Yeon Park, Robert Platt, and **Robin Walters**. A General Theory of Correct, Incorrect, and Extrinsic Equivariance. *Neural Information Processing Systems (NeurIPS)*, 2023.
- [C39] Owen Howell, David Klee, Ondrej Biza, Linfeng Zhao, and **Robin Walters**. Equivariant Single View Pose Prediction Via Induced and Restricted Representations. *Neural Information Processing Systems (NeurIPS)*, 2023.
- [C40] Jung Yeon Park, Lawson L.S. Wong[†], and **Robin Walters**[†]. Modeling Dynamics over Meshes with Gauge Equivariant Nonlinear Message Passing. *Neural Information Processing Systems (NeurIPS)*, 2023.
- [C41] Babak Esmaeili*, **Robin Walters***, Heiko Zimmermann, and Jan-Willem van de Meent. Topological Obstructions and How to Avoid Them. *Neural Information Processing Systems (NeurIPS)*, 2023.
- [C42] Dian Wang, Mingxi Jia, Xupeng Zhu, **Robin Walters**, and Robert Platt. On-Robot Learning With Equivariant Models. *Conference on Robot Learning (CoRL)*, 2022.
- [C43] Dian Wang, **Robin Walters**, and Robert Platt. SO(2)-Equivariant Reinforcement Learning. *International Conference on Learning Representations (ICLR)*, 2022.
- [C44] Rui Wang, **Robin Walters**, and Rose Yu. Approximately Equivariant Networks for Imperfectly Symmetric Dynamics. *International Conference on Machine Learning (ICML)*, 2022.
- [C45] Jung Yeon Park*, Ondrej Biza*, Linfeng Zhao, Jan-Willem van de Meent, and **Robin Walters**. Learning Symmetric Embeddings for Equivariant World Models. *International Conference on Machine Learning (ICML)*, 2022.
- [C46] Linfeng Zhao, Lingzhi Kong, **Robin Walters**, and Lawson L.S. Wong. Toward Compositional Generalization in Object-Oriented World Modeling. *International Conference on Machine Learning (ICML)*, 2022.
- [C47] Rui Wang, **Robin Walters**, and Rose Yu. Meta-learning dynamics forecasting using task inference. *Neural Information Processing Systems (NeurIPS)*, 2022.
- [C48] Bo Zhao, Nima Dehmamy, **Robin Walters**, and Rose Yu. Symmetry Teleportation for Accelerated Optimization. *Neural Information Processing Systems (NeurIPS)*, 2022.
- [C49] Haojie Huang, Dian Wang, **Robin Walters**, and Robert Platt. Equivariant Transporter Network. *Robotics: Science and Systems (RSS)*, 2022.
- [C50] Xupeng Zhu, Dian Wang, Ondrej Biza, Guanang Su, **Robin Walters**, and Robert Platt. Sample Efficient Grasp Learning Using Equivariant Models. *Robotics: Science and Systems (RSS)*, 2022.
- [C51] Dian Wang, **Robin Walters**, Xupeng Zhu, and Robert Platt. Equivariant Q-Learning in Spatial Action Spaces. *Conference on Robot Learning (CoRL)*, 2021.
- [C52] Rui Wang*, **Robin Walters***, and Rose Yu. Incorporating Symmetry into Deep Dynamics Models for Improved Generalization. *International Conference on Learning Representations (ICLR)*, 2021.

- [C53] **Robin Walters***, Jinxi Li*, and Rose Yu. Trajectory Prediction using Equivariant Continuous Convolution. *International Conference on Learning Representations (ICLR)*, 2021.
- [C54] Steven Wong, **Robin Walters**, Lejun Jiang, Tamas Molnar, Gabor Orosz, and Rose Yu. Physics-Guided Deep Learning for Traffic Forecasting using Vehicle-to-Vehicle Communication. *Learning for Dynamics and Control (L4DC)*, 2021.
- [C55] Nima Dehmamy, **Robin Walters**, Liu Yanchen, Dashun Wang, and Rose Yu. Automatic Symmetry Discovery with Lie Algebra Convolutional Network. *Neural Information Processing Systems (NeurIPS)*, 2021.

Journal Publications

- [J1] Haibo Zhao, Yu Qi, Boce Hu, Yizhe Zhu, Ziyang Chen, Heng Tian, Xupeng Zhu, Owen Howell, Haojie Huang, **Robin Walters**, Dian Wang, and Robert Platt. Generalizable hierarchical skill learning via object-centric representation. *IEEE Robotics and Automation Letters (RA-L)*, 11(6):6536–6543, 2026.
- [J2] Dian Wang, Stephen Hart, David Surovik, Tarik Kelestemur, Haojie Huang, Haibo Zhao, Mark Yeatman, Xupeng Zhu, Boce Hu, Mingxi Jia, Jiuguang Wang, **Robin Walters**, and Robert Platt. Equivariant diffusion policy for sample-efficient robotic manipulation. *International Journal of Robotics Research (IJRR)*, 2026.
- [J3] Congyue Deng, Brandon Y. Feng, Cecilia Garraffo, Alan Garbarz, **Robin Walters**, William T. Freeman, Leonidas Guibas, and Kaiming He. Denoising Hamiltonian Network for Physical Reasoning. *Transactions on Machine Learning Research (TMLR)*, 2026.
- [J4] Edward Berman, Jacob Ginesin, Marco Pacini, and **Robin Walters**. On Uncertainty Calibration for Equivariant Functions. *Transactions on Machine Learning Research (TMLR)*, 2026.
- [J5] Circe Hsu, Claire Schlesinger, Karan Mudaliar, Jordan Leung, **Robin Walters**, and Peter Schindler. FIRE-GNN: Force-Informed, Relaxed Equivariance Graph Neural Network for Rapid and Accurate Prediction of Surface Properties. *Advanced Intelligent Discovery*, pp. e202500162, 2025.
- [J6] Mingxi Jia, Haojie Huang, Zhewen Zhang, Chenghao Wang, Linfeng Zhao, Dian Wang, Jason Xinyu Liu, **Robin Walters**, Robert Platt, and Stefanie Tellex. Learning Efficient and Robust Language-conditioned Manipulation using Textual-Visual Relevancy and Equivariant Language Mapping. *IEEE Robotics and Automation Letters (RA-L)*, 2025. **Honorable Mention**.
- [J7] Boce Hu, Heng Tian, Dian Wang, Haojie Huang, Xupeng Zhu, **Robin Walters**, and Robert Platt. Push-Grasp Policy Learning Using Equivariant Models and Grasp Score Optimization. *IEEE Robotics and Automation Letters (RA-L)*, 2025.
- [J8] Sneh Pandya, Yuanyuan Yang, Nicholas Van Alfen, Jonathan Blazek, and **Robin Walters**. IAEmu: Learning Galaxy Intrinsic Alignment Correlations. *Open Journal of Astrophysics*, 2025.
- [J9] Jianke Yang, Manu Bhat, Bryan Hu, Yadi Cao, Nima Dehmamy, **Robin Walters**, and Rose Yu. Discovering Symbolic Differential Equations with Symmetry Invariants. *Transactions on Machine Learning Research (TMLR)*, 2025.
- [J10] Bo Zhao, **Robin Walters**, and Rose Yu. Symmetry in Neural Network Parameter Spaces. *Transactions on Machine Learning Research (TMLR)*, 2025.
- [J11] Sumukh Vasisht Shankar, Rui Wang, Darrel D’Souza, Jonathan P. Singer, and **Robin Walters**. Equivariant Neural Networks for Controlling Dynamic Spatial Light Modulators. *Integrating Materials and Manufacturing Innovation*, pp. 1–9, 2024.
- [J12] Niklas Smedemark-Margulies, Yunus Bicer, Elifnur Sunger, Tales Imbiriba, Eugene Tunik, Deniz Erdogmus, Mathew Yarossi, and **Robin Walters**. Fast and Expressive Gesture Recognition using a Combination-Homomorphic Electromyogram Encoder. *Transactions on Machine Learning Research (TMLR)*, 2024.

- [J13] Xupeng Zhu, Dian Wang, Guanang Su, Ondrej Biza, **Robin Walters**, and Robert Platt. On Robot Grasp Learning Using Equivariant Models. *Autonomous Robots*, 2023.
- [J14] Haojie Huang, Dian Wang, Arsh Tangri, **Robin Walters**, and Robert Platt. Leveraging Symmetries in Pick and Place. *International Journal of Robotics Research (IJRR)*, 2023.
- [J15] Ayan Chatterjee, **Robin Walters**, Zohair Shafi, Omair Shafi Ahmed, Michael Sebek, Deisy Gysi, Rose Yu, Tina Eliassi-Rad, Albert-László Barabási, and Giulia Menichetti. Improving the generalizability of protein-ligand binding predictions with AI-Bind. *Nature Communications*, 14(1):1989, 2023.
- [J16] Niklas Smedemark-Margulies*, **Robin Walters***, Heiko Zimmermann, Lucas Laird, Christian van der Loo, Neela Kaushik, Rajmonda Caceres, and Jan-Willem van de Meent. Probabilistic program inference in network-based epidemiological simulations. *PLOS Computational Biology*, 18(11):e1010591, 2022.
- [J17] Karthik Kashinath, Mustafa Mustafa, Adrian Albert, Jin Long Wu, Chiyu Jiang, Soheil Esmailzadeh, Kamyar Azizzadenesheli, Rui Wang, Alok Singh, Ashray Manepalli, Dragos Chirila, Rose Yu, **Robin Walters**, Blánaid White, Heng Xiao, Hamdi Tchelepi, Philip Marcus, Anima Anandkumar, and Prabhat. Physics-informed Machine Learning, Case Studies for Weather and Climate Modelling. *Philosophical Transactions of the Royal Society A*, 2020.

Workshop Papers and Extended Abstracts

- [W1] Lucas Laird, Bo Zhao, Rose Yu, and **Robin Walters**. Data-Free Transformer Quantization Using Parameter-Space Symmetry. *Workshop on High-dimensional Learning Dynamics (HiLD)*, 2025.
- [W2] Colin Kohler, Nathan Vaska, Ramya Muthukrishnan, Whangbong Choi, Jung Yeon Park, Justin Goodwin, Rajmonda Caceres, and **Robin Walters**. Symmetric Models for Radar Response Modeling. *Symmetry and Geometry in Neural Representations (NeurReps)*, PMLR, 2023.
- [W3] Mathilde Papillon, Mustafa Hajj, Audun Myers, Florianand Frantzen, Ghada Zamzmi, Helen Jenne, Johan Mathe, Josef Hoppe, Michael T. Schaub, Theodore Papamarkou, and others. ICML 2023 Topological Deep Learning Challenge: Design and Results. *Topological, Algebraic and Geometric Learning Workshops (TAG-ML)*, PMLR, 2023.
- [W4] Linfeng Zhao, Jung Yeon Park, Xupeng Zhu, **Robin Walters**, and Lawson L.S. Wong. SE(3) Frame Equivariance in Dynamics Modeling and Reinforcement Learning. *Workshop on Physics for Machine Learning at ICLR*, 2023.
- [W5] David Klee, Jung Yeon Park, Robert Platt, and **Robin Walters**. A Comparison of Equivariant Vision Models with ImageNet Pre-training. *Workshop on Symmetry and Geometry in Neural Representations (NeurReps) at NeurIPS*, 2023.
- [W6] Neel Sortur, Linfeng Zhao, and **Robin Walters**. Sample Efficient Modeling of Drag Coefficients for Satellites with Symmetry. *Workshop on Symmetry and Geometry in Neural Representations (NeurReps) at NeurIPS*, 2023.
- [W7] Ayan Chatterjee, **Robin Walters**, Giulia Menichetti, and Tina Eliassi-Rad. Inductive Link Prediction in Static and Temporal Graphs for Isolated Nodes. *Workshop on Temporal Graph Learning at NeurIPS*, 2023.
- [W8] Lucas Laird, **Robin Walters**, and Wolfgang Gatterbauer. Unsupervised Learning of 3-colorings using Simplicial Higher-Order Neural Networks. *Workshop on Topology, Algebra and Geometry in Machine Learning (TAG-ML) at ICML*, 2023.
- [W9] Bo Zhao, Nima Dehmamy, **Robin Walters**, and Rose Yu. Understanding Mode Connectivity via Parameter Space Symmetry. *Workshop on Unifying Representations in Neural Models (UniReps) at NeurIPS*, 2023.
- [W10] David Klee, Ondrej Biza, Robert Platt, and **Robin Walters**. Image to Icosahedral Projection for SO(3) Object Reasoning from Single-View Images. *Symmetry and Geometry in Neural Representations*

(*NeurReps*), PMLR, 2022.

- [W11] Xingyu Su, **Robin Walters**, Denis Aslangil, and Rose Yu. Forecasting Variable-density 3D Turbulent Flow. *Deep Learning for Simulation (SimDL), ICLR Workshop*, 2021.

Book Chapters

- [BC1] Rui Wang, **Robin Walters**, and Rose Yu. Physics-Guided Deep Learning for Spatiotemporal Forecasting. *Knowledge-Guided Machine Learning, Chapman and Hall/CRC*, 2023.

Publications in Pure Mathematics

- [M1] Valerio Toledano Laredo* and **Robin Walters***. On the Finkelberg–Ginzburg Mirabolic Monodromy Conjecture. *International Mathematics Research Notices*, 2024(24):14543–14575, 2024.
- [M2] Laure Flapan*, **Robin Walters***, and Xiaolei Zhao*. Period integrals and Hodge modules. arXiv:1910.00035, 2019.
- [M3] Asilata Bapat* and **Robin Walters***. The Strong Topological Monodromy Conjecture for Weyl Hyperplane Arrangements. *Mathematical Research Letters*, 24(4):947–954, 2017.
- [M4] **Robin Walters**. The Bernstein-Sato b -Function of the Space of Cyclic Pairs. *Publications of the Research Institute for Mathematical Sciences*, 51(2):273–288, 2015.
- [M5] Asilata Bapat* and **Robin Walters***. The Bernstein-Sato b -function of the Vandermonde determinant. arXiv:1503.01055, 2015.
- [M6] Colin Carroll*, Adam Jacob*, Coner Quinn*, and **Robin Walters***. The isoperimetric problem on planes with density. *Bulletin of the Australian Mathematical Society*, 78(2):177–197, 2008.

Publications in Computational Biology

- [B1] Alessandro Senes, Deborah Chadi, Peter Law, **Robin Walters**, Vikas Nanda, and William DeGrado. $E(z)$, a depth-dependent potential for assessing the energies of insertion of amino acid side-chains into membranes: derivation and applications to determining the orientation of transmembrane and interfacial helices. *Journal of Molecular Biology*, 366(2):436–448, 2007.
- [B2] Hang Yin, Joanna Slusky, Bryan Berger, **Robin Walters**, Gaston Vिलाire, Rustem Litvinov, James Lear, Gregory Caputo, Joel Bennett, and William DeGrado. Computational design of peptides that target transmembrane helices. *Science*, 315(5820):1817–1822, 2007.
- [B3] **Robin Walters** and William DeGrado. Helix-packing motifs in membrane proteins. *Proceedings of the National Academy of Sciences*, 103(37):13658–13663, 2006.

ADVISING AND MENTORING

Ph.D. students (current): Parnian Zameni, Claire Schlesinger, Yuxuan Chen, Lucas Laird, Haojie Huang, Boce Hu, Vedanth Nilabh.

Ph.D. graduates: Dian Wang (JPMorgan Ph.D. Fellow; postdoc, Stanford), Jung Yeon “John” Park (AI Researcher, Profluent), David Klee (Analog Devices), Niklas Smedemark-Margulies (Analog Devices).

Visiting Ph.D. students: Marco Pacini (University of Trento and Fondazione Bruno Kessler), Elias Nyholm (Chalmers University of Technology).

Master’s students: Neel Sortur (NSF GRFP; Peak Award), Sumukh Vasisht Shankar (Khoury Apprentice; Khoury Research Award), Yuanyuan Yang (Khoury Apprentice), Mathew Noyes, Purvik Patel.

Undergraduate students: Circe Hsu (Matterworks; Co-op; Peak Award $\times 2$), Edward Berman (Ph.D. Harvard), Luisa Li (PhD, JHU), Vedanshi Shah (Co-op).

Informal advisees: Ondrej Biza, Linfeng Zhao, Xupeng Zhu, Owen Howell, Colin Kohler.

Ph.D. defense committees: Ondrej Biza, Maria De Paolis Kaluza, Babak Esmaeili (University of Amsterdam), Sean Gunn, David Klee, Colin Kohler, Arnab Mondal (McGill), Niklas Smedemark-Margulies, Nicholas Van Alfen, Linfeng Zhao, Xupeng Zhu.

Ph.D. proposal committees: Ondrej Biza, Maria De Paolis Kaluza, Sean Gunn, David Klee, Colin Kohler, Hannah Lawrence (MIT), Sneha Pandya, Stefanos Pertigkiozoglou (University of Pennsylvania), Matt Piekenbrock, Niklas Smedemark-Margulies, Nicholas Van Alfen, Linfeng Zhao, Xupeng Zhu.

Master's thesis committees: Lingzhi Kong, Vedant Nilabh, Mathew Noyes, Andrew Sayegh, Neel Sortur, Arsh Tangri.

Qualifying exam: Owen Howell. **Ph.D. candidacy (Mathematics):** Zhaoming Li.

TEACHING

2022–present	Assistant Professor , Northeastern University CS 7180: Geometric Deep Learning (Spring 2023, Fall 2025) · CS 3180: Theory of Computation (Fall 2022, Fall 2024, Fall 2026)
2016–2020	Zelevinsky Research Instructor , Northeastern University Math 3081: Probability and Statistics · Math 1341/1342: Calculus 1 and 2 · Math 1365: Introduction to Mathematical Reasoning (4 offerings) · Putnam Exam Training (5 offerings)
2019, 2020	REU Co-organizer , Northeastern University Co-organized a summer program for 7 undergraduates and 5 graduate mentors: admissions, pairings, colloquia, and paper editing
2010–2015	Instructor and REU Mentor , University of Chicago Honors Calculus (IBL) · Introduction to Calculus · REU mentor, summers 2010–2012 · Polk Brothers Program: weekly enrichment teaching in Chicago Public Schools, 2012–2014

INVITED TALKS

Sep 2026	Mathematical Approaches to Challenges in Modern Artificial Intelligence National University of Singapore
Sep 2026	Symmetries in Neural Networks — Wallenberg Advanced Scientific Forum Rånäs Castle, Sweden
Jul 2026	SIAM Annual Meeting Special Session on Algebro-geometric Approaches to Deep Learning, Cleveland, OH
May 2026	GraphEX Symposium Dedham, MA
Dec 2025	Pushing the Limits of Equivariant Neural Networks TAG-DS Workshop, San Diego, CA
Nov 2025	How to Design Neural Networks to Understand Algebraic and Geometric Structure Institute for Computational and Experimental Research in Mathematics (ICERM), Providence, RI
Nov 2025	Equivariant Neural Networks for Dynamics and Control Woods Hole Oceanographic Institution, Woods Hole, MA
Oct 2025	Leveraging Symmetry for Learning in Physical Systems AI4Science Seminar, Georgia Tech, Atlanta, GA
Jun 2025	Pushing the Limits of Equivariant Neural Networks EquiVision Workshop, CVPR, Nashville, TN
May 2025	Pushing the Limits of Equivariant Neural Networks GRASP Seminar, University of Pennsylvania
Jan 2025	Improving Convergence and Generalization Using Parameter Symmetries AMS Special Session, Joint Mathematics Meetings, Seattle, WA

- Nov 2024 **Simulating Radar using Equivariant Graph Neural Networks**
Recent Advances in AI for National Security, Bedford, MA
- Oct 2024 **Pushing the Limits of Equivariant Neural Networks**
NeurReps Global Speaker Series, MIT
Robotics and AI Institute, Cambridge, MA
IROS Workshop on Equivariant Robotics, Abu Dhabi, UAE
- Jun 2024 **Pushing the Limits of Equivariant Neural Networks**
Computer Vision Reading Group, Caltech
AstroAI Seminar, Harvard-Smithsonian Center for Astrophysics
Symposium on Geometry Processing Graduate School, MIT
- May 2024 **Improving Convergence and Generalization Using Parameter Symmetries**
ICLR Oral, Vienna, Austria
- Apr 2024 **Everyday Robotics**
AI in Action, Northeastern University
- Feb 2024 **Symmetry in Deep Neural Networks**
Open Neighborhood Seminar, Department of Mathematics, Harvard University
- Nov 2023 **The Strengths and Limitations of Equivariant Neural Networks**
AI/ML in Physics, Georgia Tech, Atlanta, GA
- Oct 2023 **The Strengths and Limitations of Equivariant Neural Networks**
Widely Applied Mathematics Seminar, Harvard SEAS
- Sep 2023 **Equivariant Neural Networks for Dynamics and Control**
Department of Mechanical and Aerospace Engineering Colloquium, Rutgers University
- Aug 2023 **Incorporating Symmetry into Deep Dynamics Models · The Strengths and Limitations of Equivariant Neural Networks**
IAIFI Summer School Tutorial and IAIFI Summer Workshop, Northeastern University
Graph Exploitation Symposium, Dedham, MA
- Mar 2023 **Symmetry in Deep Neural Networks · The Strengths and Limitations of Equivariant Neural Networks**
Colloquium, Michigan State University Department of Mathematics
Machine Learning Seminar, Department of Mathematics, UMass Amherst

Earlier talks (2015–2022). MIT-IBM Watson AI Lab (2022) · Physics-informed Machine Learning Series, CNLS, Los Alamos National Laboratory (2022) · NSF Workshop on Foundations of Machine Learning for Scientific Discovery, Tyson’s Corner, VA (2022) · Mathematics in Imaging, Data, and Optimization Seminar, Rensselaer Polytechnic Institute (2022) · Geometry and Physics Seminar, Boston University (2022) · Flatiron Institute (2021) · AMLab Seminar, University of Amsterdam (2021) · AI Seminar, ECE, UC San Diego (2021) · SciML Webinar, Carnegie Mellon University (2021) · INFORMS Session on Spatiotemporal Data (2021) · Computational Sciences Department, Princeton Plasma Physics Laboratory (2021) · MIT Lincoln Laboratory (2021) · Deep Learning Reading Group, Duke University Department of Mathematics (2021) · Spring Eastern AMS Sectional Meeting, Special Session on Arrangements of Hypersurfaces (2018) · Representations and Lie Theory Seminar, Ohio State University (2017) · Algebra and Geometry Seminar, Sapienza University of Rome (2016) · Géométrie Algébrique en Liberté XXIII and XXIV, Leuven and Nesin Mathematics Village (2015, 2016) · Algebraic Geometry Seminar, Purdue University (2015) · Lie Seminar, UC Riverside (2015).

PROFESSIONAL SERVICE

Program Chair Learning on Graphs (LoG) 2026

Workshop and conference organization: Topology, Algebra and Geometry in Data Science × Boston
Symmetry Day 2026 · Geometry-grounded Representation Learning and Generative Modeling (GRaM),

ICML 2024 and ICLR 2026 · Geometric and Algebraic Structure in Robot Learning, RSS 2024 · Boston Symmetry Day (Spring 2023, Fall 2023, Fall 2024, Spring 2025) · Symmetries in Robot Learning, RSS 2023.

Area Chair: ICLR (2024–2026), Learning on Graphs (2024–2026), ICML (2026), NeurIPS (2026), GRaM Workshop (2024).

Journal and conference reviewing: ICLR, ICML, NeurIPS, KDD (2020–2025) · JMLR ($\times 3$), IEEE TPAMI, Applied and Computational Harmonic Analysis, Foundations of Computational Mathematics, Compositio Mathematica, Pure and Applied Mathematics Quarterly, Journal of Singularities, Transportation Research Part C · NeurReps and AI4Science workshops at NeurIPS.

Grant review: NSF, ELLIS, ISF.

Northeastern service: Faculty Hiring Committee, Khoury College (2022–2024) · Research Computing Advisory Committee, Khoury College (2024–2026).

Outreach: Organizer, Northeastern–O’Bryant School of Math and Science Machine Learning Program: four-session program for 60 Boston public school students (2022, 2024) · Problem writer, Bridge to Calculus, Boston high school program (2017–2019).

Seminar organization: Geometry, Physics, and Representation Theory Seminar, Northeastern (2017–2020) · Pick My Brain Seminar, Northeastern, co-founder (2016–2020) · Postdoctoral Research Seminar, Northeastern, founder (2015–2016) · Graduate Seminar on Hodge Modules, Northeastern and MIT (2016).